LOAD|BY|TON.store — Operations Handbook

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The one place for every step and command needed to run the live site. Written for a non-technical owner. Tick the boxes (☐) as you go. **Never paste passwords, keys or tokens into chat, email or screenshots.**

1. The system at a glance

Piece	What it does	Where
Website + app	Runs the product	One AWS server (EC2), region **eu-north-1 (Stockholm)**
Server ID	Used to connect	`i-0f878ed97ff158915`
Code	The program	GitHub `loadbyton-byte/Loadbyton-OS`, branch `main`
Internet entrance	Domain, security, tunnel to server	Cloudflare (domain `loadbyton.store`)
Database	All customer data	Supabase (PostgreSQL, Europe)
Files + backups	Uploaded documents, nightly backups	Cloudflare R2 (two buckets: files, backups)
Rate limits	Stops abuse	Upstash (Europe)
Errors	Alerts when code breaks	Sentry
Payments	Card payments	Stripe (test mode now)
Email	Invitations, receipts	Resend (domain verified)

Web addresses: `https://www.loadbyton.store` (site) · `https://admin.loadbyton.store` (admin) · `https://www.loadbyton.store/healthz` (should show `{"ok":true}`).

Server folder: `/opt/loadbyton`. Settings file: `/opt/loadbyton/.env.production` (secret, never share).

2. Golden rules

1. **Backup before any change.** The deploy script does it automatically.
2. **Change one thing at a time**, then check `/healthz`.
3. **Secrets live only in** `.env.production` on the server, your password manager and the provider dashboards.
4. If a secret was ever pasted somewhere public: **rotate it** (create a new one, update the server, delete the old).
5. Money: never switch to live payment keys until the launch checklist (section 13) is fully ticked.
6. Watch the AWS bill weekly (budget about US$30/month against the US$100 credit).

3. Connect to the server (every time)

On your Chromebook Linux terminal:

```
aws ssm start-session --target i-0f878ed97ff158915 --region eu-north-1
sudo -i
cd /opt/loadbyton
```

You should see a prompt ending `/opt/loadbyton#`. To leave: type `exit` twice.

If the command fails, the AWS keys on the Chromebook may have changed: run `aws configure` (region `eu-north-1`) and use a **new** access key from AWS IAM.

4. Check everything is healthy

```
curl -s https://www.loadbyton.store/healthz
bash infra/aws/compose.sh ps
bash infra/aws/compose.sh logs --tail=40 web
bash infra/aws/compose.sh logs --tail=40 worker
bash infra/aws/configure-env.sh check
```

- `healthz` must print `{"ok":true}`.
- `ps` must show web, worker, scanner, tunnel all running/healthy.
- `check` lists which settings are set or EMPTY (never prints secrets).

5. Deploy a new version (manual, always works)

Use this after changes are **merged into `main` on GitHub**. Find the newest commit address with `git rev-parse origin/main` (a 40-character code).

☐ 1. Connect (section 3), then `cd /opt/loadbyton`. ☐ 2. Back up first: `bash infra/aws/backup.sh` (must end `backup OK: uploaded and found in R2`). ☐ 3. Deploy (replace COMMIT with the 40-character code):

```
git fetch origin
git checkout --quiet --detach COMMIT
bash infra/aws/compose.sh build
bash infra/aws/compose.sh run --rm --no-deps web npm run migrate
bash infra/aws/compose.sh up -d --remove-orphans
```

☐ 4. Wait a minute, then check:

```
bash infra/aws/compose.sh ps
curl -s https://www.loadbyton.store/healthz
curl -s https://www.loadbyton.store/readyz
```

Success: all containers healthy, `{"ok":true}` and `{"ok":true,"workers":"ok"}`.

Roll back (use the commit you were on before; `git rev-parse HEAD` before deploying shows it):

```
git checkout --quiet --detach OLD_COMMIT
bash infra/aws/compose.sh build
bash infra/aws/compose.sh up -d --remove-orphans
```

Database changes are not reversed by a rollback.

About the scripted deploy (`infra/aws/deploy.sh`): it also runs `readiness --strict`, which refuses until every launch gate in the readiness register is signed off. Until then, use the manual steps above. The automatic GitHub deploy has the same limit.

6. Automatic deploys from GitHub (one-time setup)

After this, every merge to `main` tests and deploys by itself (watch it in GitHub → Actions).

☐ 1. In GitHub repo → **Settings** → **Secrets and variables** → **Actions** → **Variables**, add:

Variable	Value
`AWS_REGION`	`eu-north-1`
`AWS_DEPLOY_ROLE_ARN`	output `github_deploy_role_arn` from Terraform (section 12)
`AWS_INSTANCE_ID`	`i-0f878ed97ff158915`

☐ 2. **Settings** → **Environments** → **New environment** named `production`. ☐ 3. Merge a small change to `main` and watch **Actions** turn green. ☐ 4. Check `/healthz`.

7. Settings (secrets and configuration)

Change a setting safely:

```
cd /opt/loadbyton
bash infra/aws/configure-env.sh
```

It asks one question at a time. **Press Enter to keep the current value.** Long values (database address) are typed in pieces: host, project ref, password. After changes, restart:

```
bash infra/aws/compose.sh up -d
bash infra/aws/compose.sh exec web npm run preflight
```

`preflight` must say `Deployment setting validation passed.`

Database certificate error (`SELF_SIGNED_CERT_IN_CHAIN`): run `bash infra/aws/supabase-ca.sh`, then restart.

Upstash message `[ratelimit] upstash unavailable... fetch failed`: the app still works with a built-in counter, but fix it: in the Upstash dashboard copy the REST URL and token of the **Europe** database, run `configure-env.sh`, enter them, restart.

Two keys to keep in a password manager (generated by the script, cannot be recovered): `BACKUP_ENCRYPTION_KEY` and `MFA_ENCRYPTION_KEY`. Without them backups and 2-step logins cannot be recovered.

8. Backups and restore

Nightly backup at 02:30 server time → encrypted `.lbt` file → R2 backups bucket.

One-time setup (already done):

```
bash infra/aws/backup.sh setup
bash infra/aws/backup.sh install-cron
```

Run a backup now and check: `bash infra/aws/backup.sh` → must end `backup OK: uploaded and found in R2`. Log: `/var/log/loadbyton-backup.log`.

Restore drill (do before launch, then quarterly) — see `docs/RUNBOOKS.md`:

☐ 1. Create a **new, empty** Supabase project (never use the live one) and a different private storage bucket. ☐ 2. Download the newest `.lbt` from the R2 backups bucket. ☐ 3. Set `RESTORE_DATABASE_URL`, `RESTORE_CONFIRM_ISOLATED=true`, the backup key, and the different bucket; run `npm run restore -- <file.lbt>`. ☐ 4. Run `npm run verify` against the restored database: it must print `All invariants hold.` ☐ 5. Log in to the restored copy, open a document, open the audit trail. Write down how long it took.

Never point restore at the live database.

9. Admin users and protection

Create an admin (the password is typed hidden, never put in the command):

```
cd /opt/loadbyton
read -rp "Admin email: " E
read -rp "Name: " N
read -rsp "Password (12+ characters): " P; echo
bash infra/aws/compose.sh exec -T web npm run create-staff -- "$E" "$N" "$P" commercial_admin
```

It prints a secret for your authenticator app (Google Authenticator, etc.). Add it to the app immediately. Roles: `commercial_admin`, `support`, `engineer`, `auditor`.

Protect the admin address with Cloudflare Access:

☐ Cloudflare → **Zero Trust** → **Access** → **Applications** → **Add** → **Self-hosted** ☐ Hostname `admin.loadbyton.store` ☐ Policy: **Allow**, rule **Emails**, only your email address ☐ Test in a private window: a Cloudflare code screen must appear first.

10. Email (Resend)

Already verified. To change the sender or key: run `configure-env.sh`, answer the Resend questions (key `re_...`, sender `noreply@loadbyton.store`), restart. Test by inviting a team member to your own address. Check the spam

folder.

11. Payments (Stripe)

Today: **test mode**. Webhook address: `https://www.loadbyton.store/webhooks/payments/stripe` (5 events).

Go live — only when section 13 is complete:

☐ 1. Stripe dashboard → switch to **Live mode**; complete business verification. ☐ 2. Create a **live** webhook with the same address and the same 5 events; copy its signing secret (`whsec_...`). ☐ 3. Server: `bash infra/aws/configure-env.sh` → enter the live secret key (`sk_live_...`) and the new signing secret. ☐ 4. Edit `.env.production` and **delete the line** `ALLOW_TEST_PAYMENT_KEYS=true`. ☐ 5. `bash infra/aws/compose.sh up -d` then `... exec web npm run preflight`. ☐ 6. Make one small real purchase, confirm the order shows paid, then refund it.

12. AWS and Terraform commands (on the Chromebook)

Run from the repo folder on the Chromebook (not on the server): `cd Loadbyton-OS/infra/aws`.

Task	Command
See what would change	`terraform plan`
Apply changes	`terraform apply` (read the plan, type `yes`)
Show outputs (server ID, deploy role)	`terraform output`
Check AWS login works	`aws sts get-caller-identity --region eu-north-1`

Important: the AWS account only allows region **eu-north-1**. Other regions are blocked. **Never run `terraform destroy`** — it deletes the server.

Reboot the server if ever needed: AWS console → EC2 → instance → Instance state → Reboot. The containers restart by themselves.

13. Launch checklist (before real customers)

☐ Site and `/healthz` OK · ☐ Admin login works with 2-step code ☐ Cloudflare Access on `admin.loadbyton.store` ☐ Nightly backup ran and `backup OK` seen · ☐ **Restore drill done and recorded** ☐ Email invitation test received · ☐ Stripe test payment and webhook = 200 ☐ `preflight` passes · ☐ `readiness --strict` passes (deploy script runs it) ☐ `CHECKOUT_TAX_NOTE` wording approved by your accountant ☐ Privacy policy, terms, 90-day retention reviewed (`LEGAL_REVIEWED=true`) ☐ Sentry receiving events · ☐ AWS budget alert set ☐ Stripe live keys in, `ALLOW_TEST_PAYMENT_KEYS` removed, one real purchase + refund tested ☐ Old Mumbai Supabase/Upstash projects deleted

14. Troubleshooting

Symptom	Likely cause	Fix
Site down, `healthz` fails	Container stopped	`bash infra/aws/compose.sh up -d`, then logs (section 4)
`ECONNREFUSED 127.0.0.1:5432`	Database address wrong or broken by paste	Re-enter with `configure-env.sh` (3 short pieces)
`SELF_SIGNED_CERT_IN_CHAIN`	Missing database CA	`bash infra/aws/supabase-ca.sh`
Deploy: `backup.sh: No such file`	Server on old commit	Fix in section 5
Deploy: checkout would overwrite `backup.sh`	File not executable	`chmod +x infra/aws/backup.sh && git add infra/aws/backup.sh`
`UNHEALTHY... rolling back`	New version failed	Read last 25 lines; old version stays
Upstash `fetch failed`	Wrong/old Upstash URL/token	Section 7
Admin cannot log in	Wrong 2-step code or no admin	Create admin (section 9)
Stripe event not 200	Wrong signing secret	Re-copy `whsec_...`, `configure-env.sh`, restart
Invitation email missing	Resend domain/key	Check Resend domain is Verified; section 10
AWS `AccessDenied` / SCP error	Wrong region	Always use `eu-north-1`

15. Where to ask for help

Give: the **exact command** you ran and the **last 25 lines** of output. Never include passwords, keys, tokens or whole `.env.production` contents.

Source files for everything here: `infra/aws/` (scripts), `docs/RUNBOOKS.md`, `docs/MONITORING.md`, `docs/DEPLOY_AWS_LEAN.md`.